Individual Contribution in Capstone Project

Project Name: Demand Prediction by SKU - Sales Data Collection and EDA

Group No:

Sponsor: Deloitte

Sno	Name	Email Id	PGID	Percentage Contributed to the Project	Tasks done (Mention the tasks done by respective team member in the project)
1	**Sai Prashanth Sarathy**	Sai_Sarathy_ampba2026S@isb.edu	12510081	**20%**	• **Lead Platform Architecture & Vercel Deployment:** Architected serverless cloud deployment on Vercel with 100% localhost parity (commits 441fa1e, fe13e8d, e82675c), APICS replenishment smoothing telemetry, and automated routing. • **End-to-End Data Pipeline & Ingestion:** Built canonical automated data pipeline (src/data/orchestrator.py, ingestion.py, cleaning.py), transforming 82,034 raw ERP records into the 8,951-row canonical weekly data mart. • **System Architecture & Governance:** Structured the modular 7-layer decoupled software hierarchy and centralized YAML configuration governance (config/data_config.yaml, model_config.yaml, business_config.yaml). • **Documentation & Reporting:** Primary author of Weekly Reports Wk 02, Wk 03, and co-author of Wk 07; compiled system ARCHITECTURE.md and forensic audit documentation.
2	**R Arvindshankar**	Arvind_Shankar_ampba2026S@isb.edu	12510053	**20%**	• **Multi-Echelon Network Modeling:** Structured the 3-echelon supply chain network (National → South Region → 5 Regional Depots → 154 SKUs) in src/hierarchy/aggregation.py. • **Hierarchical Mathematical Reconciliation:** Implemented src/hierarchy/reconciliation.py containing Bottom-Up, Top-Down, and Minimum Trace (MinT) optimal combination reconciliation with Ledoit-Wolf shrinkage covariance estimation, guaranteeing zero reconciliation mismatches (+4.8% accuracy gain). • **Econometric Variance Decomposition:** Built classical additive time-series decomposition module analyzing Trend, Seasonality, and Residual variance shares across SKU tiers. • **Reporting & Formulation Defense:** Co-author of Weekly Report Wk 07; drafted formal mathematical formulation proofs, structural summation matrices (S matrix), and mid-review defense guide.

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3	**Hariharan N**	Hariharan_N_ampba2026S@isb.edu	12510102	**20%**	• **Analytics Cockpit & UI/UX Architecture:** Designed and coded the 7-module interactive intelligence dashboard (dashboard.py, src/dashboard_intelligence.py), providing seamless navigation across Executive Overview, Data Quality Radar, Cleaning Studio, Deep-Dive Forecasting, Inventory Optimizer, and What-If Simulator. • **Interactive Visualization Engine:** Built the dynamic Plotly charting suite featuring dual light/dark high-contrast theme responsiveness, confidence interval fan charts, and multi-dimensional hierarchy filtering across 154 SKUs and branches. • **Initial Data Extraction & CRM Validation:** Spearheaded initial raw ERP data extraction from manufacturing sponsor, validating regional sales office codes, customer tier classifications, and dealer master records. • **Reporting & Slide Deck Design:** Primary author of Weekly Report Wk 04; co-author of Wk 01; designed capstone presentation slide visual templates and UI style guide.
4	**Vedika Sharma**	Vedika_Sharma_ampba2026S@isb.edu	12510071	**20%**	• **Causal Feature Engineering:** Developed src/features/engineering.py implementing strictly shifted autoregressive lags (1–52W), rolling demand momentum statistics, and Fourier cyclical features with zero temporal data leakage. • **Machine Learning Tournament Engine:** Implemented recursive walk-forward forecasting (predict_recursive) for tree ensembles (Random Forest, Gradient Boosting, XGBoost, LightGBM) in src/forecasting/ml_models.py. • **Walk-Forward Backtesting & Leakage Audit:** Engineered expanding-window chronological backtesting (src/validation/backtesting.py) and 4-tier automated leakage auditor (src/validation/leakage_audit.py), achieving 100% clean audit pass. • **Reporting & Model Governance:** Primary author of Weekly Report Wk 05; compiled MODEL_CARD.md, LEAKAGE_AUDIT.md, and auditable Champion/Challenger model registry.

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5	**Vani Som**	Vani_Som_ampba2026S@isb.edu	12510023	**20%**	• **Probabilistic Inventory Engine:** Programmed src/inventory/inventory_engine.py converting forecast distributions into actionable replenishment policies: Lead Time Demand (LTD), Safety Stock (SS) with service level calibration (90–98%), dynamic Reorder Points (ROP), and Monte Carlo stockout simulations. • **Conformal Prediction Intervals:** Engineered empirical P10, P50, and P90 uncertainty intervals in src/uncertainty/prediction_intervals.py. • **Scenario Simulation & Operational Telemetry Radar:** Built What-If simulator (src/scenarios/simulator.py) and statistical drift monitor (src/monitoring/drift.py, alerts.py) utilizing Population Stability Index (PSI), KS tests, and APICS Tracking Signals. • **Reporting & Business ROI:** Primary author of Weekly Report Wk 06; co-author of Wk 01; authored business impact, stockout risk reduction, and working capital optimization case studies.